

Attention-Weighted Retrieval-Augmented Data Engineering for ADR Severity Classification

David Guo and Nishant Vishwamitra and Kim-Kwang Raymond Choo

Department of Information Systems and Cybersecurity

The University of Texas at San Antonio

david.guo@utsa.edu, nishant.vishwamitra@utsa.edu, raymond.choo@utsa.edu

Abstract

Adverse drug reaction (ADR) severity prediction is a key task in pharmacovigilance and post-market drug safety surveillance, but is often challenged by noisy and heterogeneous clinical data. We propose a retrieval-augmented learning framework that leverages historical FAERS reports to construct evidence-enriched training instances for ADR severity prediction. Each ADR query, including patient demographics, drug exposure, and indications, is encoded using a fine-tuned BGE-M3 embedding model. We retrieve the top five relevant cases using BM25 lexical search and the top five semantically similar cases using dense vector search, and fuse them via an attention-based aggregation mechanism to produce retrieval-aware representations. These representations are evaluated using machine learning, deep learning, transformer, and fine-tuned large language model approaches for binary severity prediction. Results show that a tree-based ensemble achieves the best performance with an AUC of 82.3%, outperforming the strongest deep learning baseline by 2.3 percentage points and fine-tuned LLMs by 17 percentage points. Overall, the findings demonstrate that retrieval-enhanced data preprocessing can improve representation quality for ADR severity prediction, while the proposed upstream data engineering approach enables classical ensemble methods to remain highly effective for large-scale pharmacovigilance risk prediction¹.

1 Introduction

Adverse drug reaction (ADR) poses a significant risk to patient safety and have been reported as the third leading cause of death following heart disease and cancer (Makary and Daniel, 2016). As the number of approved and experimental drugs continues to increase, early and accurate prediction of ADR outcomes has become increasingly critical

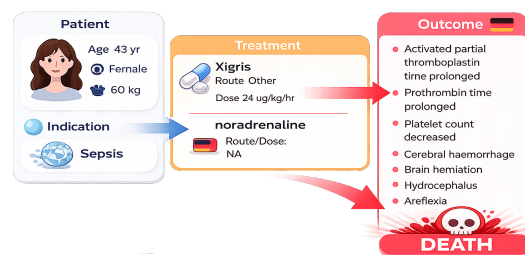


Figure 1: Illustrative Example of an ADR Severity Outcome

for minimizing patient risk and improving clinical outcomes (Mullard, 2026). The U.S. Food and Drug Administration Adverse Event Reporting System (FAERS) serves as one of the largest repositories of post-market drug safety reports, containing millions of spontaneously reported adverse events collected over more than two decades. It records more than 1.25 million serious ADR cases in 2022 including nearly 175,000 fatalities accessed in 2024 (Kommu et al., 2024). Accurate prediction of ADR severity from these reports can assist regulators, healthcare providers, and pharmaceutical manufacturers in prioritizing safety signals and allocating investigative resources. Figure 1 depicts an example ADR prediction task, consisting of the ADR structured label (e.g., “DEATH”) and unstructured narratives describing the label in preferred clinical terminology, given a patient’s information and treatment.

Recent advances in natural language processing have led to widespread adoption of transformer-based architectures and large language models (LLMs) for biomedical text understanding. Models such as BERT, Llama, and Qwen have demonstrated impressive performance across a range of clinical and biomedical tasks (Deimazar and Sheikhtaheri, 2023; Golder et al., 2025). However, ADR severity prediction presents several challenges that limit the effectiveness of purely para-

¹The project repository is publicly available in GitHub repository: <https://github.com/nexusmed-ai/EMNLP2026>

metric models. First, severe ADR cases are relatively rare, resulting in highly imbalanced training data. Second, ADR narratives are mixed with noisy and heterogeneous information, making it difficult for language models to capture subtle severity-related patterns. Third, clinically relevant evidence frequently exists in historical reports rather than within the current case itself, requiring models to leverage external pharmacovigilance knowledge.

To address these challenges, we propose an attention-weighted retrieval-augmented representation learning framework for ADR severity prediction. Instead of relying solely on the information contained in a single ADR report, our approach retrieves semantically and lexically similar historical cases from a FAERS repository spanning 24 years of pharmacovigilance data (Robertson and Zaragoza, 2009; Wang et al., 2025; Lamsiyah et al., 2023). Specifically, each query report is matched with historical cases using both BM25 lexical retrieval and dense vector retrieval based on a fine-tuned BGE-M3 embedding model². Retrieved cases are then integrated through an attention-weighted aggregation mechanism that dynamically estimates the relevance of each neighbor and constructs a retrieval-enhanced contextual representation for downstream prediction. Next, traditional tree-based ensemble methods remain competitive or superior when predictive signals are dominated by structured attributes and retrieval-derived features, allowing them to leverage heterogeneous information more efficiently than large neural architectures.

The proposed framework is evaluated on approximately 800,000 retrieval-enhanced ADR instances derived from FAERS quarterly reports and historical pharmacovigilance data. We compare traditional machine learning methods (Random Forest, XGBoost), deep neural networks (Convolutional Neural Network (CNN), long short-term memory networks (LSTMs), and bidirectional LSTMs (BiLSTMs)), transformer-based architectures, and fine-tuned large language models. Results show that retrieval-enhanced representations significantly improve predictive performance and generalization to future reports. Notably, tree-based ensembles consistently outperform transformer and LLM-based models, suggesting that retrieval quality and feature engineering are more critical than model complexity for ADR severity prediction.

This work makes three primary contributions as follows:

1. we introduce an attention-weighted retrieval framework that integrates lexical and semantic evidence from historical pharmacovigilance reports to enrich ADR representations.
2. we provide a large-scale empirical study comparing traditional ML, DL, and LLMs under a unified retrieval-augmented learning setting
3. we demonstrate through out-of-time evaluation that retrieval-enhanced representations improve robustness and predictive performance for ADR severity prediction, highlighting the value of historical pharmacovigilance knowledge for biomedical NLP applications.

2 Related Work

2.1 ADR Prediction and Pharmacovigilance

The increasing availability of large-scale pharmacovigilance databases such as FAERS has driven extensive research on ADR prediction. Prior work has applied supervised learning methods using FAERS and external biomedical data, including Random Forest and deep learning models that integrate demographic, molecular, and biological features to predict ADRs, demonstrating the effectiveness of machine learning at scale (Farnoush et al., 2024). Deep neural networks combining toxicogenomic data with FAERS reports have also shown promising performance across multiple ADR categories (Mohsen et al., 2021).

In contrast, ADR severity prediction has received comparatively less attention, with most studies focusing on structured and unstructured data mining and information retrievals efforts. For example, structured approaches using FDA labeling and MedDRA terminology define seriousness categories for pharmacovigilance analysis (Wu et al., 2019), while attention-based models have been used to improve serious ADR detection using NLP and demographic features in FAERS (Aldughayfiq et al., 2024). Other approaches estimate severity from social media word embeddings and network propagation methods at a small scale, showing correlations with real-world clinical outcomes (Lavertu et al., 2021).

2.2 Deep Learning and Transformer-Based Approaches

Advances in biomedical NLP have accelerated the adoption of deep learning architectures for pharma-

²<https://huggingface.co/BAAI/bge-m3>

covigilance tasks. CNNs, LSTMs, and BiLSTMs have been widely used for ADR extraction, classification, and signal detection (Miranda, 2018; Gupta et al., 2018; Zhang and Geng, 2019). More recently, transformer-based models and LLMs have demonstrated strong performance on biomedical text understanding tasks due to their ability to capture contextual semantic information (Lee et al., 2020; Alsentzer et al., 2019; Gu et al., 2022; Singhal et al., 2025; Wang et al., 2024). Nevertheless, ADR severity prediction remains challenging because pharmacovigilance narratives are often sparse, noisy, and highly imbalanced. Existing transformer-based approaches typically rely on information contained within individual reports and do not explicitly leverage historical pharmacovigilance knowledge.

2.3 Retrieval-Augmented Learning

Retrieval-augmented methods have recently emerged as an effective strategy for incorporating external knowledge into prediction tasks by dynamically retrieving relevant evidence and conditioning models on non-parametric memory. Prior work in dense retrieval enabled neural document retrieval for open-domain tasks (Karpukhin et al., 2020), while retrieval-augmented language models integrated retrieved passages into downstream generation and classification (Lewis et al., 2021; Guu et al., 2020). Nearest-neighbor language modeling further improved generalization by leveraging example-based reasoning through explicit retrieval of similar training instances (Khandelwal et al., 2020). Collectively, these approaches demonstrate the benefit of combining parametric models with external memory, though their application to pharmacovigilance remains limited, particularly in leveraging historical ADR reports for severity prediction.

To address this gap, we propose an attention-weighted retrieval-augmented framework that integrates lexical, dense, and similarity-based retrieval over two decades of FAERS data. Unlike prior work relying on single-instance narratives, our method systematically incorporates historical pharmacovigilance evidence and compares traditional ML, DL, and LLM-based approaches within a unified retrieval-enhanced learning setting.

2.4 Finetuning Language Model Approaches

Advances in language model fine-tuning, particularly BERT and its biomedical variants, have improved the representation of ADR-related seman-

tics and enhanced downstream prediction tasks (Guo and Choo, 2025). LLMs can also be leveraged to generate structured and unstructured narratives in preferred clinic terminology (Zhou et al., 2024). This shift enables more effective exploitation of rich, unstructured healthcare data, despite the inherent challenges posed by complex domain knowledge and heterogeneous data structures.

3 Our Approach

3.1 ADR Data Preparation

FAERS is released as quarterly compressed archives containing multiple plain-text files along with accompanying processing guidelines. Each release includes five core tables that can be linked via a unique caseID to reconstruct individual ADR reports.

We design an extraction, transformation, and loading (ETL) pipeline consisting of five steps illustrated as **Figure 2**. First, data extraction unpacks each quarterly FAERS archive and parses raw text files according to FDA specifications. Second, data transformation aligns and joins records across five tables using caseID to construct unified patient-centric ADR cases. Third, outcome processing maps the OUTC field to seven categorical outcomes; approximately 26% of reports are missing outcome labels. FAERS data are divided into three temporal splits: quarterly releases from 2004Q1–2024Q3 are used to construct the hybrid retrieval pipelines, 2024Q4–2025Q1 are used for training and validation, and 2025Q2 is reserved as an out-of-time (OOT) test set. To improve exposure to diverse drug–reaction patterns, ADR cases with missing outcome codes are included during embedding model and LLM fine-tuning. However, for fair evaluation of structured classification performance, we restrict metrics to cases with non-missing outcome labels.

A binary severity label is further derived by assigning the life-threatening class (DE/LT) when at least one life-threatening outcome is present among multiple reported outcomes, resulting in a highly imbalanced distribution (79% negative and 21% positive). Fourth, missing values in non-critical fields (e.g., dose frequency) are imputed as NA while drug name and indication are required to be non-missing. Finally, each ADR case is formatted as an instruction–outcome pair in JSONL and stored in Parquet format to enable efficient BM25 and vector indexing (See **Appendix**).

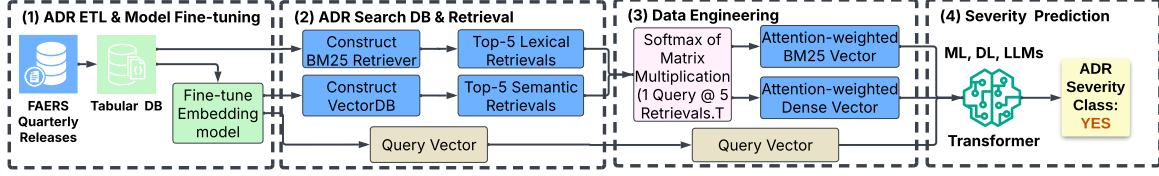


Figure 2: Overview of the Retrieval-Augmented ADR Severity Prediction Framework

3.2 ADR Instruction, Uni_code, and Severity Class

FAERS tabular data are aggregated from quarterly releases dating back to 2004Q1 up to 2025Q2, where each ADR case includes patient demographics, drug administration details, medication indications, represented by MedDRA preferred terms, and outcome labels. Each case is formatted as an instruction–outcome pair, with the instruction composed of 3 components: patient demographics (age, gender, weight), drug administration details (drug name, dose, route, frequency), and medication indications. To preserve the original column-wise relationships while supporting retrieval and generation tasks, all instruction–outcome pairs are stored in line-delimited JSON (JSONL) format, with each record containing a self-contained structured representation. The JSONL representation is the query format for both BM25 and vector hybrid retrieval stage. An illustrative example of instruction is shown in **Appendix 7.1**.

Each ADR case outcome consists of two components: an unstructured narrative formed by concatenated MedDRA Preferred Terms (PTs) and a structured outcome label representing the most severe reported event. Only structured outcome codes are used to construct the binary severity target (YES/NO), as summarized in Table 1. When multiple outcomes are present, the highest-severity label is assigned as the uni_code. Both the instruction and outcome are stored in a line-delimited JSON (JSONL) format to ensure consistency, as illustrated in **Appendix 7.2**.

3.3 Fine-tuning Embedding Model

High-quality retrieval is critical for retrieval-augmented generation (RAG) because retrieved evidence directly influences downstream ADR prediction. Rather than fine-tuning large language models, which is computationally expensive and requires frequent retraining as new FAERS data become available, we fine-tuned a domain-specific embedding model for efficient and scalable re-

Code	Severity	Description
DE	YES	Death
LT	YES	Life-Threatening
HO	NO	Hospitalization-Initial or Prolonged
DS	NO	Disability
CA	NO	Congenital Anomaly
RI	NO	Required Intervention to Prevent Permanent Impairment/Damage
OT	NO	Other Serious (Important Medical Event)

Table 1: Definition of Binary Severity Classification from Seven Structured ADR Outcomes

trieval. We selected BAAI/BGE-M3³ as the base model due to its strong retrieval performance and fine-tuned it on 500,000 ADR instruction–outcome pairs using Matryoshka representation learning (Schroff et al., 2015; Hoffer and Ailon, 2015; Krašniković et al., 2025; Ng et al.).

Model performance was evaluated on a held-out validation set using Top-K Hit Rate, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG). The best-performing checkpoint was selected for subsequent vector database construction. We further compared the fine-tuned ADR-specific embedding model with three MedEmbed variants designed for biomedical and clinical text (Balachandran, 2024). As reported in **Table 2**, the fine-tuned BGE-M3 model substantially outperforms both the base BGE-M3 model and the MedEmbed baselines across all retrieval metrics. The fine-tuned embedding model is publicly available on Hugging Face at https://huggingface.co/nexusmed-ai/adr_bge_m3_embedding.

model	hit@1	hit@3	hit@5	mrr@5	ndcg@5
bge_m3 FT	0.81	0.97	1.00	0.89	0.92
bge_m3 Base	0.24	0.56	0.86	0.44	0.54
med_small	0.29	0.63	0.89	0.49	0.59
med_base	0.22	0.56	0.87	0.44	0.54
med_large	0.26	0.60	0.88	0.47	0.57

Table 2: Finetuning Bge-m3 with Matryoshka Representation Learning

³<https://huggingface.co/BAAI/Bge-m3>

3.4 Model Training Dataset

For each ADR report, the query narrative was encoded into a 1,024-dimensional embedding using the fine-tuned BGE-M3 model. The top five BM25-retrieved cases and top five dense-retrieved cases were encoded and aggregated separately using similarity-based attention weights, producing one BM25 context embedding and one dense context embedding. These three 1,024-dimensional vectors—the query embedding, BM25 context embedding, and dense context embedding—were stacked to form a (3 X 1024) retrieval-aware tensor for downstream ADR severity prediction.

ML models use a flattened 3,072-dimensional feature vectors, whereas CNN, LSTM, BiLSTM, and Transformer models consume the full $3 \times 1,024$ tensor. Fine-tuned LLMs operate on the original text instructions and retrieved cases without explicit embedding inputs.

3.5 Construction of Retrieval Database

To enable efficient similarity search over millions of ADR embeddings, we store all query representations in a LanceDB vector database using an IVF-HNSW-PQ approximate nearest-neighbor index with 256 partitions and 128 sub-vectors⁴, providing a balance between search efficiency and recall. At query time, the top-5 most similar ADR cases are retrieved based on cosine similarity, which reduces to a dot product due to unit-normalized BGE-M3 embeddings. The resulting ranked neighbors are then passed to downstream fusion and re-ranking stages for retrieval-augmented prediction.

In parallel, we constructed a BM25 lexical retriever using the built-in tokenizer provided by the Python bm25s library for both indexing and querying to ensure consistent tokenization throughout the retrieval pipeline. At inference time, the top-5 matching ADR cases are retrieved based on lexical similarity. This stage complements dense retrieval by providing precise matching of patient demographics, drug names, and indication information.

3.6 Attention-Weighted Data Engineering

With three data channels derived from the two retrieval pipelines (BM25 and dense retrieval) and the query embedding from Box 2 of Figure 2, we further process the retrieved information in three

steps: 1) Similarity computation - a matrix multiplication $q \times R^\top$ between the query embedding $q \in R^{1 \times 1024}$ and the 5 retrieved case embeddings $R \in R^{5 \times 1024}$ to yields a 1×5 vector of cosine similarities. 2) Attention weighting - softmax is applied to these 5 similarities, producing attention weights as

$$w_i = \frac{\exp(\text{sim}(q, r_i)\tau)}{\sum_{j=1}^5 \exp(\text{sim}(q, r_j)/\tau)} \quad (1)$$

where τ is the temperature parameter controlling the concentration of the attention distribution. A smaller τ produces a more concentrated distribution, placing greater emphasis on the most similar retrievals, whereas a larger τ produces a smoother distribution (Hinton et al., 2015; Guo et al., 2017). In this study, we set $\tau=1$ to maintain a relatively smooth weighting across the five retrieved cases. 3) Context aggregation - each of the top-5 embeddings is multiplied by its attention weight, and the weighted vectors are summed into a single aggregated context vector as $c = \sum_{i=1}^5 w_i \cdot r_i$. where $c \in R^{1 \times 1024}$ represents the attention-weighted retrieval context used in subsequent prediction.

3.7 LLM Finetuning and Inference

We fine-tune two open-source LLMs of different sizes - Qwen-3.5-0.8B (text only)⁵ and Llama-3.2-3B⁶ on 80,000 FAERS reports from training set with binary labels. Both 4-bit quantized based models using QLoRA with LoRA adapters (rank 16, alpha 32, dropout 0.05) on the attention layers. Only the LoRA adapters and classification head are updated, while the backbone remains frozen for memory-efficient training. The models are optimized using an imbalance-aware cross-entropy objective over the combined query and retrieval context. Fine-tuning was conducted on a single NVIDIA V100 (32GB) GPU, requiring approximately 9 hours for Qwen-3.5-0.8B and 52 hours for Llama-3.2-3B. During inference, the final binary severity label is determined by majority voting over predictions on the query case and its 10 retrieved neighbors.

4 Evaluation

4.1 Comparative Model Evaluation on OOT

Across all models under imbalanced data, traditional machine learning methods demonstrate the

⁵<https://huggingface.co/principled-intelligence/Qwen3.5-0.8B-text-only>

⁶<https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>

⁴<https://docs.lancedb.com/indexing/vector-index>

strongest and most stable discriminative performance prioritizing F1. Among them, Random Forest achieves the best F1-score (0.513) with strong AUC (0.800), while XGBoost obtains the highest AUC (0.803), indicating robust ranking ability under class imbalance. Deep learning models, including Transformer, CNN, BiLSTM, and LSTM, show moderate but consistently lower performance compared to tree-based methods. The Transformer performs best within this group (F1 0.476, AUC 0.799), while CNN and BiLSTM are slightly weaker, and LSTM performs worst, suggesting limited benefit from sequential modeling in this high-dimensional and noisy setting.

In contrast, fine-tuned LLMs show consistent improvements on F1 and Recall over their base models but remain less competitive overall. Qwen3.5-0.8B improves across all metrics compared to its baseline, while Llama3.2-3B increases recall (0.833) and F1 (0.278) but sacrifices precision (0.167), indicating strong positive-class bias. Overall, both LLM families underperformed classical machine learning models in terms of F1 and AUC, highlighting the robustness of tree-based ensemble models for ADR severity prediction under substantial class imbalance (0.79 negative vs. 0.21 positive).

Model	Accu.	F1	Rec.	Prec.	AUC
XGBoost	0.819	0.489	0.525	0.506	0.803
RF	0.828	0.513	0.491	0.501	0.800
Transformer	0.811	0.476	0.568	0.518	0.799
CNN	0.810	0.472	0.529	0.499	0.796
BiLSTM	0.793	0.436	0.545	0.485	0.787
LSTM	0.777	0.410	0.560	0.473	0.783
Qwen3.5-.8B FT	0.841	0.445	0.350	0.610	0.650
Qwen3.5-.8B Base	0.808	0.098	0.059	0.274	0.513
Llama3.2-3B FT	0.252	0.278	0.833	0.167	0.482
Llama3.2-3B base	0.816	0.061	0.035	0.262	0.507

Table 3: Evaluation of ML, DL, and LLM Performance

4.2 Majority Vote and Close-Sourced LLM Inference Baselines

To further evaluate alternative inference strategies, we included majority voting and GPT-5-mini inference as additional baselines on a subset of the OOT test set. Majority voting was implemented as a frequency-ranked ensemble of BM25 and dense-retrieval labels. For each case, the top-5 outcome labels retrieved by BM25 and the top-5 labels retrieved by dense embedding similarity were pooled into a single candidate set and ranked by vote count. We considered two variants: unweighted majority voting, which ranked labels by their raw vote counts, and score-weighted majority voting, in

Inference Setting	F1	Precision	Recall
Majority vote (unweighted)	0.137	0.773	0.075
Majority vote (score-weighted)	0.180	0.674	0.104
Zero-shot	0.134	0.268	0.090
Prompt w/ Context (RAG)	0.242	0.491	0.161
Proposed model	0.513	0.506	0.525

Table 4: Comparison of majority voting and GPT-5-mini inference strategies on a subset of the OOT test set. The performance of proposed model are the best scores from ensemble model from Table 3

which each vote was weighted by its corresponding retrieval similarity score.

We also evaluated the closed-source GPT-5-mini model under zero-shot and RAG settings for the same binary classification task of predicting whether an ADR case was likely to result in a life-threatening outcome. GPT-5-mini was accessed through OpenAI’s Batch API with minimal reasoning effort and prompted to act as a pharmacovigilance expert, interpret the available clinical information—including patient demographics, drug exposure, route and dose, and treatment indication—and return a single JSON prediction ("life_threatening": 0 or "life_threatening": 1). Responses were parsed using strict JSON parsing. For zero-shot inference, the model received only the query case and generated its prediction based on the information contained in that case. For RAG inference, the query case was augmented with 10 retrieved reference cases as in-context evidence: the top-5 BM25 and top-5 dense-vector nearest neighbors, each annotated with its ground-truth life-threatening label (0/1). These labeled cases were appended to the query as an evidence block. The model was instructed to use the retrieved cases as supporting evidence and base its prediction solely on the query case and retrieved evidence, without relying on external knowledge.

We evaluated these four inference strategies, unweighted majority voting, score-weighted majority voting, GPT-5-mini zero-shot inference, and GPT-5-mini RAG inference—and compared their performance with that of the proposed model. The results are summarized in Table 4.

The results indicate that direct LLM inference and majority-voting strategies achieve substantially lower F1 scores than the proposed retrieval-augmented fine-tuned ensemble model (F1 = 0.513). Among the evaluated LLM-based approaches, prompt-based inference with 10 retrieved cases achieved the highest F1 score (0.242), suggesting that retrieved contextual evidence can improve inference performance over zero-shot

prompting. Score-weighted majority voting also improved over unweighted majority voting, increasing the F1 score from 0.137 to 0.180. Nevertheless, all baseline strategies remained substantially below the proposed model, demonstrating the benefit of jointly leveraging retrieval-augmented evidence and supervised fine-tuning for the target prediction task. This comparison is included in the appendix to provide a more comprehensive evaluation of alternative inference strategies.

4.3 Stability of Ensemble Model Performance

The proposed F1-weighted ensemble achieved robust and stable performance across 5-fold cross-validation (CV) on the training dataset. Thresholds were optimized to maximize the F1 score, balancing precision and recall for the imbalanced ADR severity prediction task. The precision–recall curves demonstrate strong performance under class imbalance, with average precision (AP) values ranging from 0.603 to 0.623 and a mean PR-AUC of 0.613 ± 0.009 , substantially exceeding the positive-class prevalence of 0.21. The consistently high AP scores and low variance across folds indicate stable identification of severe ADR cases despite the skewed class distribution (Figure 3). Overall, the ensemble model achieves a favorable balance between precision and recall while maintaining robust ranking performance, suggesting that retrieval-enhanced representations provide complementary predictive signals that generalize well across heterogeneous validation folds.

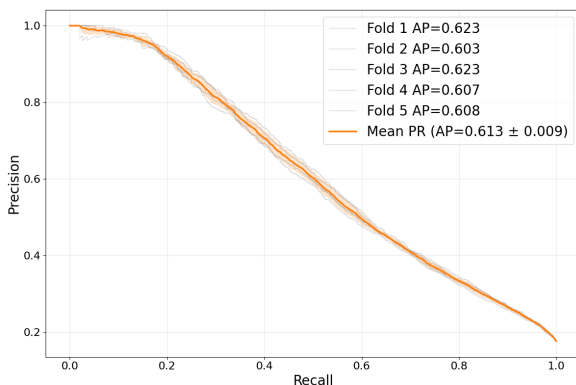


Figure 3: Imbalance-Aware Evaluation Using Precision–Recall AUC with 5-Fold Cross-Validation on the Validation Set

4.4 Ablation Study Individual Retrieval Inputs

An ablation study was conducted on the held-out OOT dataset to assess the contribution of three

input vector representations under severe class imbalance, using F1-score, ROC-AUC, and PR-AUC as the primary evaluation metrics (Table 5). The BM25 representation performs worst overall, achieving high precision (0.588) but extremely low recall (0.076), leading to poor F1 (0.134), ROC-AUC (0.580), and PR-AUC (0.275), indicating limited generalization and poor coverage of positive cases. In contrast, the query and dense representations show substantially stronger and comparable performance, with F1-scores of 0.435 and 0.436, ROC-AUC values of 0.768 and 0.763, and PR-AUC scores of 0.488 and 0.494, respectively, demonstrating improved discrimination under class imbalance. The weighted ensemble combines the three input vectors by weighting their predicted probabilities according to validation F1 scores, achieving the most robust balanced performance with the highest ROC-AUC (0.773), best F1-score (0.458), and strongest PR-AUC (0.506). These results show that integrating complementary retrieval signals consistently improves both ranking quality and precision–recall trade-offs in the OOT setting.

Vector	Accu.	Prec.	Rec.	F1	AUC	PR
Ensemble	0.807	0.447	0.469	0.458	0.773	0.506
Query	0.829	0.510	0.379	0.435	0.768	0.488
Dense	0.812	0.453	0.421	0.436	0.763	0.494
BM25	0.831	0.588	0.076	0.134	0.580	0.275

Table 5: Ablation Study of Three Input Representations. Ensemble Vector represents the element-wise average of the three input vectors: the Query, Dense Retrieval, and BM25 Retrieval vectors.

5 Conclusion

We presented an attention-weighted retrieval-augmented framework for ADR severity prediction using 20 years of FAERS data. By combining lexical, dense, and similarity-based retrieval signals, the framework encodes historical ADR evidence into structured representations aiming to improve scalability, latency, computation efficiency, and deployment cost. Across ML, DL, and LLM evaluations, we demonstrated that retrieval-enhanced feature engineering consistently improves performance under noisy and imbalanced data. Notably, well-designed retrieval and feature construction enable classical tree-based ensembles to outperform more complex neural and LLM-based approaches, highlighting the importance of retrieval quality and feature engineering for practical pharmacovigilance systems.

6 Limitations

Despite the strong empirical performance of the proposed framework, several limitations remain. First, the study relies on voluntarily submitted FAERS reports, which are inherently noisy, incomplete, and subject to substantial reporting bias. The quality and level of detail vary considerably across reporters, and many reports contain missing, inconsistent, or inaccurate information. Furthermore, FAERS submissions generally do not undergo systematic expert clinical review or validation prior to inclusion in the database, meaning that reported drug–event associations may not reflect confirmed causal relationships. As a result, reporting artifacts, and unverified adverse events can introduce noise into both the retrieval corpus and training data. These limitations could affect the reliability of learned patterns and reduce the generalizability of downstream predictions, particularly for rare events, newly approved drugs, and underrepresented patient populations.

Second, while retrieval augmentation improves predictive performance, it introduces dependence on the quality of the retrieved evidence. The model assumes that the retrieved historical cases are relevant and representative of the query case; however, retrieval errors can propagate directly into downstream predictions. For example, BM25 retrieval may overemphasize lexical overlap while missing semantically relevant cases, whereas dense retrieval may return conceptually similar but clinically unrelated reports. In addition, historical biases and reporting patterns present in FAERS can be amplified when frequently retrieved cases disproportionately influence model decisions. As a result, retrieval quality becomes a critical factor affecting model robustness, generalizability, and interpretability, particularly when applied to emerging drugs, rare adverse events, or previously unseen patient populations.

Third, the current framework aggregates historical cases using retrieval-based representations without explicitly incorporating temporal information. Although the FAERS corpus spans more than two decades, retrieved cases are ranked primarily by semantic similarity, ignoring their reporting year and quarter. This limitation is important because drug–event relationships and severe outcomes may evolve over time due to changes in prescribing patterns, regulatory actions, and emerging safety signals. As a result, older cases may bias retrieval

and weaken the model’s ability to capture recent severity trends. Future work could address this by incorporating timestamp-aware retrieval, such as encoding year and quarter information or applying time-sensitive weighting during ranking.

Fourth, the temperature parameter (τ) in Equation 1 used in the retrieval-attention mechanism was fixed rather than extensively tuned. The temperature controls the sharpness of the softmax distribution over the cosine similarities of the retrieved cases and therefore directly influences how much the model emphasizes the most similar retrievals versus aggregating information across multiple retrieved cases. In this study, we set $\tau=1$ to maintain a relatively smooth attention distribution. Although this choice reduces the risk of overemphasizing a single potentially noisy retrieval, alternative values of τ may yield different performance, particularly when the similarity scores of the retrieved cases are closely distributed. A systematic sensitivity analysis or validation-based tuning of τ could potentially improve retrieval weighting and provide a more robust assessment of the contribution of individual retrievals. Future work should investigate adaptive or learnable temperature parameters that can dynamically calibrate retrieval attention based on query characteristics and retrieval confidence.

Fifth, although we evaluated a broad range of model families, including ML, DL, and LLMs, the LLM experiments are limited to relatively small open-source models (0.8B and 3B parameters). These models may lack the capacity and domain-specific knowledge required for complex pharmacovigilance reasoning and may not fully represent the capabilities of larger state-of-the-art LLMs. Consequently, the observed performance gap between LLMs and traditional ML approaches should be interpreted with caution, as larger LLMs or models further adapted to biomedical and pharmacovigilance data may achieve stronger performance, albeit at substantially higher computational and inference costs.

Finally, interpretability of the combined retrieval-and-learning pipeline remains partially constrained, particularly in how individual retrieved instances contribute to final predictions. Addressing these limitations through improved data curation, temporal modeling, and more interpretable retrieval attribution represents an important direction for future work.

Acknowledgments

Computational resources were provided by The University of Texas at San Antonio Arc High-Performance Computing (HPC) cluster. This research was also partially supported by the Exploratory and Predictive Analytics on Adverse Drug Reactions project (MDE260011), funded by the National Science Foundation Advanced Cyberinfrastructure Awards to the Texas Advanced Computing Center.

References

- Bader Aldughayfiq, Hisham Allahem, Ayman Mohamed Mostafa, Mohammed Alnusayri, and Mohamed Ezz. 2024. [Layer-Weighted Attention and Ascending Feature Selection: An Approach for Seriousness Level Prediction Using the FDA Adverse Event Reporting System](#). *Applied Sciences*, 14(8):3280.
- Emily Alsentzer, John R. Murphy, Willie Boag, Wei-Hung Weng, Di Jin, Tristan Naumann, and Matthew B. A. McDermott. 2019. [Publicly Available Clinical BERT Embeddings](#). *arXiv preprint*. ArXiv:1904.03323 [cs.CL].
- Abhinand Balachandran. 2024. [Medembed: Medical-focused embedding models](#).
- Ghasem Deimazar and Abbas Sheikhtaheri. 2023. [Machine learning models to detect and predict patient safety events using electronic health records: A systematic review](#). *International Journal of Medical Informatics*, 180:105246.
- Alireza Farnoush, Zahra Sedighi-Maman, Behnam Rasoolian, Jonathan J. Heath, and Banafsheh Fallah. 2024. [Prediction of adverse drug reactions using demographic and non-clinical drug characteristics in FAERS data](#). *Scientific Reports*, 14:23636.
- Su Golder, Dongfang Xu, Karen O'Connor, Yunwen Wang, Mahak Batra, and Graciela Gonzalez Hernandez. 2025. [Leveraging Natural Language Processing and Machine Learning Methods for Adverse Drug Event Detection in Electronic Health/Medical Records: A Scoping Review](#). *Drug Safety*, 48(4):321–337.
- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2022. [Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing](#). *ACM Transactions on Computing for Healthcare*, 3(1):1–23. ArXiv:2007.15779 [cs.CL].
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. [On Calibration of Modern Neural Networks](#). *arXiv preprint*. ArXiv:1706.04599 [cs.LG].
- David Guo and Kim-Kwang Raymond Choo. 2025. [Applications of Federated Large Language Model for Adverse Drug Reactions Prediction: Scoping Review](#). *Journal of Medical Internet Research*, 27(1):e68291.
- Shashank Gupta, Sachin Pawar, Nitin Ramrakhiyani, Girish Keshav Palshikar, and Vasudeva Varma. 2018. [Semi-Supervised Recurrent Neural Network for Adverse Drug Reaction mention extraction](#). *BMC bioinformatics*, 19(Suppl 8):212.
- Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. [REALM: Retrieval-Augmented Language Model Pre-Training](#). *arXiv preprint*. ArXiv:2002.08909 [cs.CL].
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. [Distilling the Knowledge in a Neural Network](#). *arXiv preprint*. ArXiv:1503.02531 [stat.ML].
- Elad Hoffer and Nir Ailon. 2015. [Deep Metric Learning Using Triplet Network](#). In *Similarity-Based Pattern Recognition*, pages 84–92, Cham. Springer International Publishing.
- Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. [Dense Passage Retrieval for Open-Domain Question Answering](#). *arXiv preprint*. ArXiv:2004.04906 [cs.CL].
- Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. 2020. [Generalization through Memorization: Nearest Neighbor Language Models](#). *arXiv preprint*. ArXiv:1911.00172 [cs.CL].
- Sharath Kommu, Christopher Carter, and Philip Whitfield. 2024. [Adverse Drug Reactions](#). In *StatPearls [Internet]*. StatPearls Publishing.
- Ceca Kraišniković, Robert Harb, Markus Plass, Wael Al Zoughbi, Andreas Holzinger, and Heimo Müller. 2025. [Fine-tuning language model embeddings to reveal domain knowledge: An explainable artificial intelligence perspective on medical decision making](#). *Engineering Applications of Artificial Intelligence*, 139:109561.
- Salima Lamsiyah, Abdelkader El Mahdaouy, Said Ouatiq El Alaoui, and Bernard Espinasse. 2023. [Un-supervised query-focused multi-document summarization based on transfer learning from sentence embedding models, BM25 model, and maximal marginal relevance criterion](#). *Journal of Ambient Intelligence and Humanized Computing*, 14(3):1401–1418.
- Adam Lavertu, Tymor Hamamsy, and Russ B. Altman. 2021. [Quantifying the Severity of Adverse Drug Reactions Using Social Media: Network Analysis](#). *Journal of Medical Internet Research*, 23(10):e27714.
- Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo

- Kang. 2020. [BioBERT: a pre-trained biomedical language representation model for biomedical text mining](#). *Bioinformatics*, 36(4):1234–1240. ArXiv:1901.08746 [cs.CL].
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. [Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks](#). *arXiv preprint*. ArXiv:2005.11401 [cs.CL].
- Martin A. Makary and Michael Daniel. 2016. [Medical error—the third leading cause of death in the US](#). *BMJ*, 353:i2139. Publisher: British Medical Journal Publishing Group Section: Analysis.
- Diego Saldana Miranda. 2018. [Automated Detection of Adverse Drug Reactions in the Biomedical Literature Using Convolutional Neural Networks and Biomedical Word Embeddings](#). *arXiv preprint*. ArXiv:1804.09148 [cs.CL].
- Attayeb Mohsen, Lokesh P. Tripathi, and Kenji Mizuguchi. 2021. [Deep Learning Prediction of Adverse Drug Reactions Using Open TG-GATEs and FAERS Databases](#). *Frontiers in Drug Discovery*, 1:768792. ArXiv:2010.05411 [q-bio.GN].
- Asher Mullard. 2026. [2025 FDA approvals](#). *Nature Reviews Drug Discovery*, 25(2):81–87.
- Karen Ka Yan Ng, Izuki Matsuba, and Peter Chengming Zhang. [RAG in health care: A novel framework for improving communication and decision-making by addressing LLM limitations](#). 2(1):AIra2400380. Publisher: Massachusetts Medical Society.
- Stephen Robertson and Hugo Zaragoza. 2009. [The Probabilistic Relevance Framework: BM25 and Beyond](#). *Foundations and Trends® in Information Retrieval*, 3(4):333–389. Publisher: Now Publishers, Inc.
- Florian Schroff, Dmitry Kalenichenko, and James Philbin. 2015. [FaceNet: A Unified Embedding for Face Recognition and Clustering](#). In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–823. ArXiv:1503.03832 [cs].
- Karan Singhal, Tao Tu, Juraj Gottweis, Rory Sayres, Ellery Wulczyn, Mohamed Amin, Le Hou, Kevin Clark, Stephen R. Pfohl, Heather Cole-Lewis, Darlene Neal, Qazi Mamunur Rashid, Mike Schaeckermann, Amy Wang, Dev Dash, Jonathan H. Chen, Nigam H. Shah, Sami Lachgar, Philip Andrew Mansfield, and 16 others. 2025. [Toward expert-level medical question answering with large language models](#). *Nature Medicine*, 31(3):943–950.
- Chong Wang, Mengyao Li, Junjun He, Zhongruo Wang, Erfan Darzi, Zan Chen, Jin Ye, Tianbin Li, Yanzhou Su, Jing Ke, Kaili Qu, Shuxin Li, Yi Yu, Pietro Liò, Tianyun Wang, Yu Guang Wang, and Yiqing Shen. 2024. [A Survey for Large Language Models in Biomedicine](#). *arXiv preprint*. ArXiv:2409.00133 [cs.CL].
- Mengzhao Wang, Boyu Tan, Yunjun Gao, Hai Jin, Yingfeng Zhang, Xiangyu Ke, Xiaoliang Xu, and Yifan Zhu. 2025. [Balancing the Blend: An Experimental Analysis of Trade-offs in Hybrid Search](#). *arXiv preprint*. ArXiv:2508.01405 [cs].
- Leihong Wu, Taylor Ingle, Zhichao Liu, Anna Zhao-Wong, Stephen Harris, Shraddha Thakkar, Guangxu Zhou, Junshuang Yang, Joshua Xu, Darshan Mehta, Weigong Ge, Weida Tong, and Hong Fang. 2019. [Study of serious adverse drug reactions using FDA-approved drug labeling and MedDRA](#). *BMC bioinformatics*, 20(Suppl 2):97.
- Min Zhang and Guohua Geng. 2019. [Adverse Drug Event Detection Using a Weakly Supervised Convolutional Neural Network and Recurrent Neural Network Model](#). *Information*, 10(9):276.
- Hongjian Zhou, Fenglin Liu, Boyang Gu, Xinyu Zou, Jinfa Huang, Jing Wu, Yiru Li, Sam S. Chen, Peilin Zhou, Junling Liu, Yining Hua, Chengfeng Mao, Chenyu You, Xian Wu, Yefeng Zheng, Lei Clifton, Zheng Li, Jiebo Luo, and David A. Clifton. 2024. [A Survey of Large Language Models in Medicine: Progress, Application, and Challenge](#). *arXiv preprint*. ArXiv:2311.05112 [cs.CL].

7 Appendix

7.1 Construction of ADR Instruction in JSONL

```
{ "patient": "{ \"age\": \"43.0\", \"age_cod\": \"yr\", \"gndr_cod\": \"f\", \"wt\": \"60.0\", \"wt_cod\": \"kg\" }\", \"treatment\": \"\"drugname\": \"xigris\", \"route\": \"other\", \"dose\": \"24 ug/kg/hr/other\"; { \"drugname\": \"noradrenaline\", \"route\": nan, \"dose\": nan }\", \"indi_pt\": \"sepsis\" }
```

7.2 Construction of ADR Outcome in JSONL

```
{ \"pt\": \"activated partial thromboplastin time prolonged; areflexia; brain herniation; cerebral haemorrhage; hydrocephalus; platelet count decreased; prothrombin time prolonged; pupil fixed\", \"uni_code\": \"DE\" }
```